

An Overview of ASPA, ASRA, and BAR-SAV, and Their Performance Modeling

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IETF Drafts:

ASPA: <https://datatracker.ietf.org/doc/draft-ietf-sidrops-aspa-verification/>

ASRA: <https://datatracker.ietf.org/doc/draft-sriram-sidrops-asra-verification/>

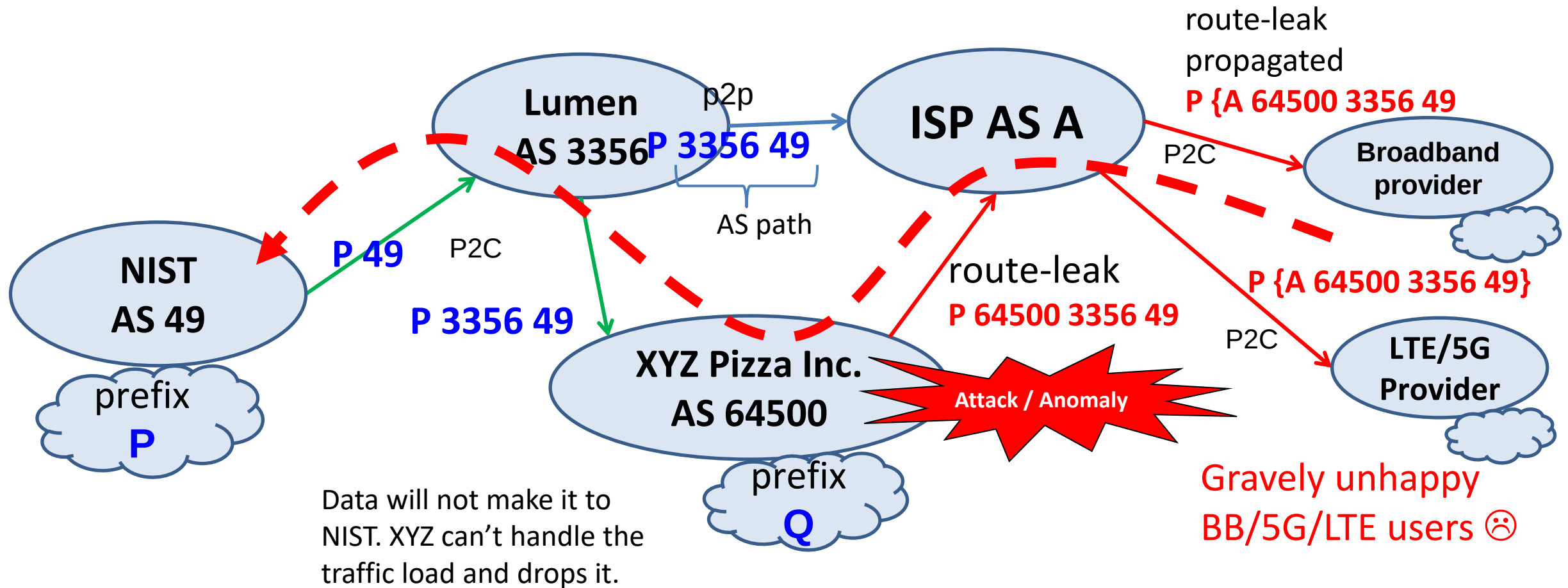
BAR-SAV: <https://datatracker.ietf.org/doc/draft-ietf-sidrops-bar-sav/>

NIST project: <https://www.nist.gov/programs-projects/robust-inter-domain-routing>

IETF 126, Vienna, Austria, July 2026

Route Leak

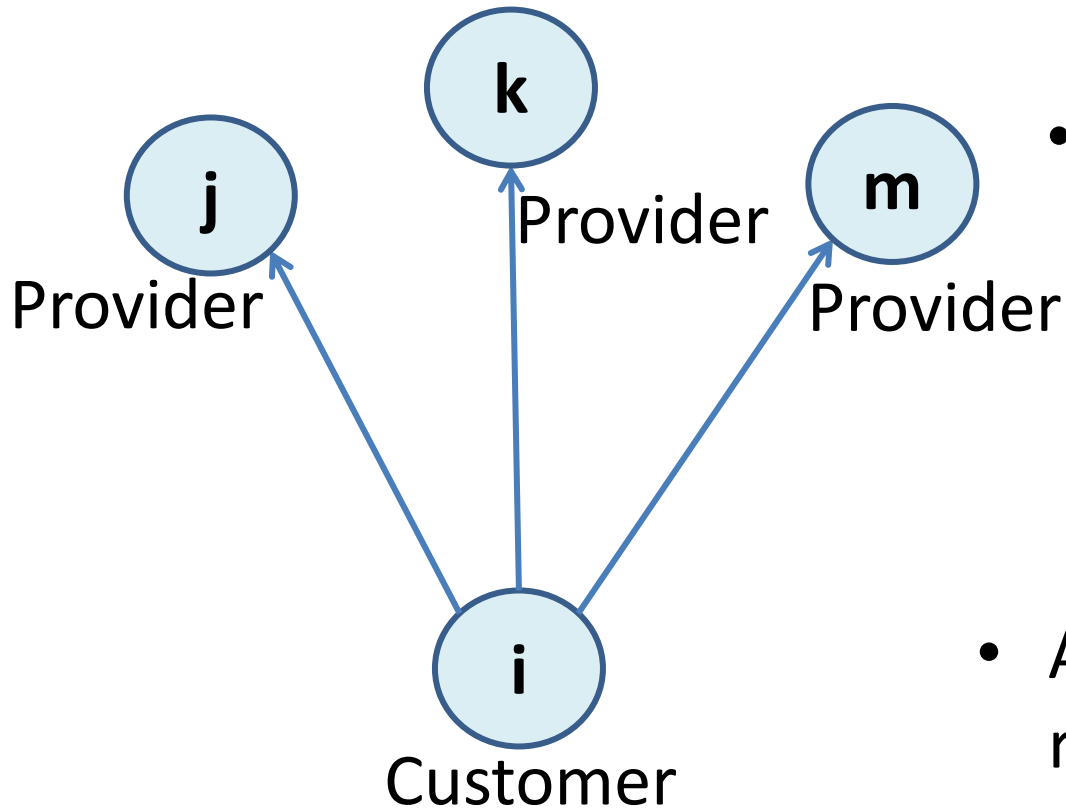
← - Anomalous data flow path



Note: This is only an illustration.

ASPA: Autonomous System Provider Authorization

RPKI ASPA Object



Example:

- ASPA: AS i, {AS j, AS k, AS m}
transit providers

AS i signs an ASPA object in the RPKI to attest that AS j, AS k, and AS m are transit providers

- ASPAs are registered/stored in the RPKI repositories

For details of ASPA registration requirements, see Section 4 in <https://datatracker.ietf.org/doc/html/draft-ietf-sidrops-aspa-verification>

ASPA Hop Check Function

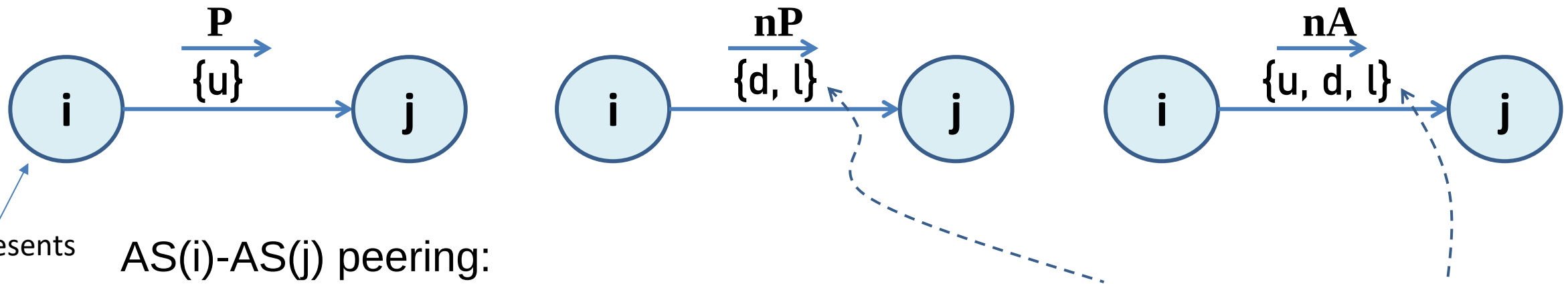
Definition:

$$\text{hop}(\text{AS}(i), \text{AS}(j)) = \begin{cases} \mathbf{P} & \text{if AS}(i) \text{ attests AS}(j) \text{ is a provider} \\ \mathbf{nP} & \text{if AS}(i) \text{ attests AS}(j) \text{ is not a provider} \\ \mathbf{nA} & \text{if AS}(i) \text{ does not have an ASPA} \end{cases}$$

P: Provider

nP: not Provider

nA: no Attestation



Represents
AS(i)

AS(i)-AS(j) peering:

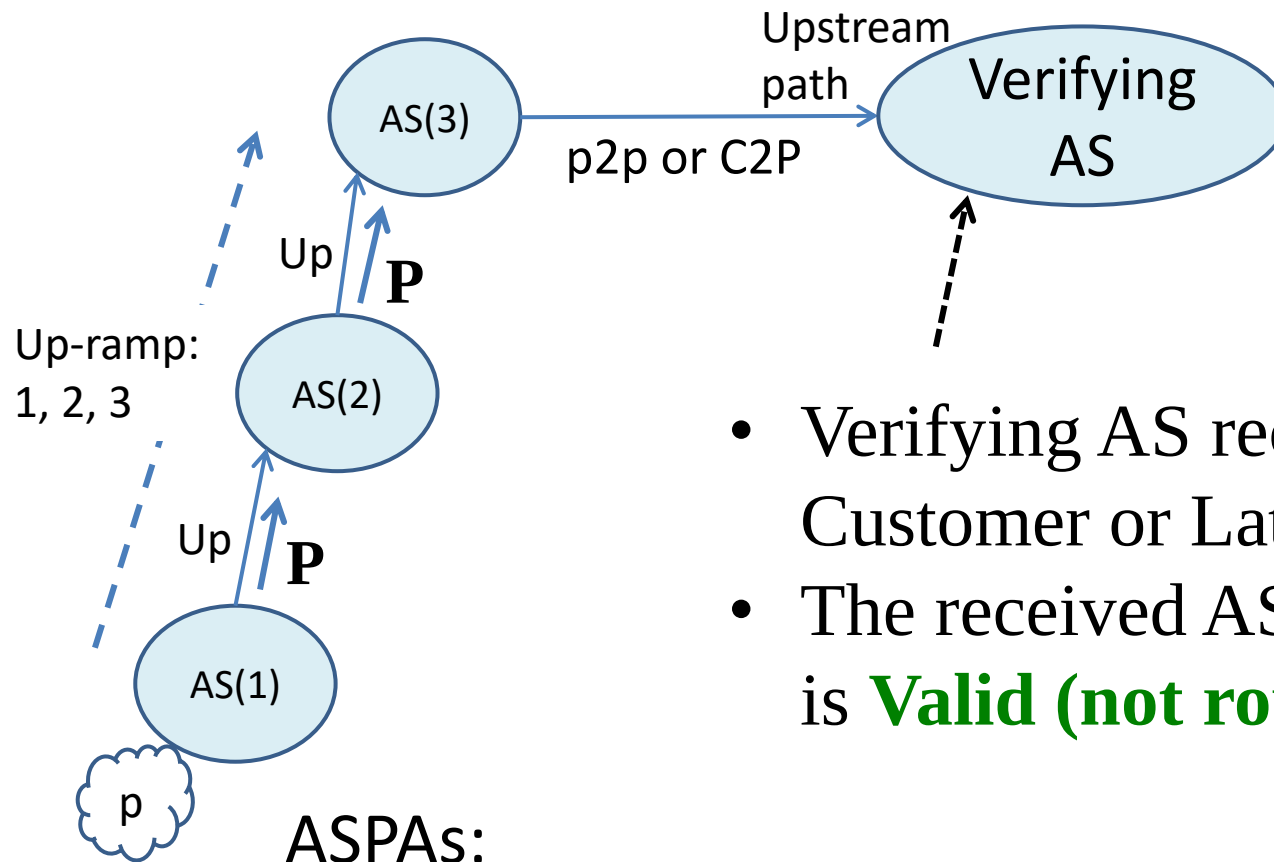
u = Up (customer to provider (C2P))

d = Down (provider to customer (P2C))

l = Lateral (peer to peer (p2p))

allowed peering relations

Example when Upstream AS Path is **Valid**



ASPAs:

AS(1), {AS(2)}

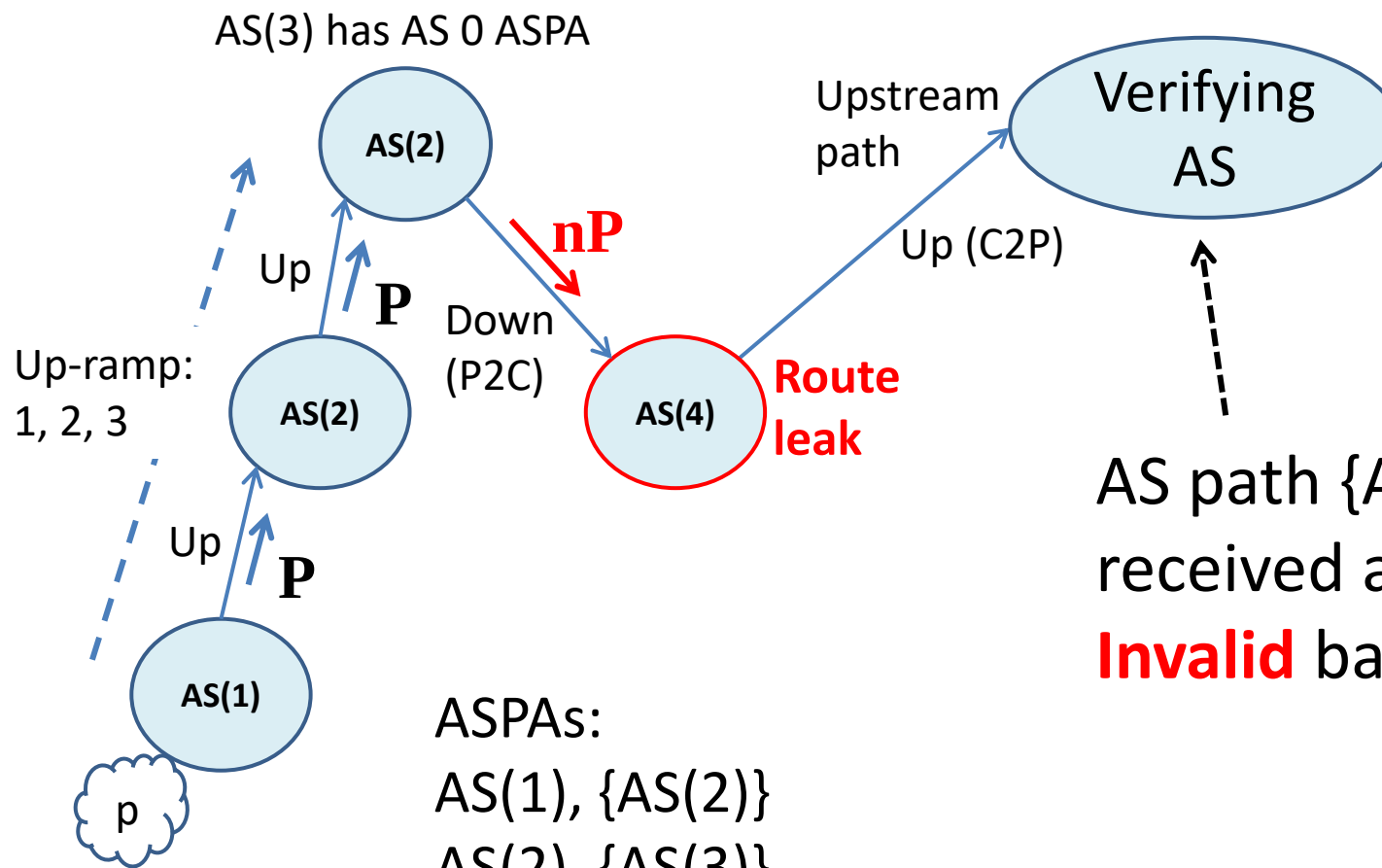
AS(2), {AS(3)}

- Verifying AS receives the BGP route from a Customer or Lateral Peer;
- The received AS path {AS(3) AS(2) AS(1)} is **Valid (not route leak)**

ASPA hop check:
P: Provider
nP: not Provider
nA: no Attestation

AS(3) may or may not have an ASPA. The Verifying AS has local knowledge that AS(3) is its customer or lateral peer.

Example when Upstream AS Path is **Invalid**



ASPA hop check:
P: Provider
nP: not Provider
nA: no Attestation

AS path {AS(4) AS(3) AS(2) AS(1)}
received at the Verifying AS is
Invalid based on ASPAs

ASPAs:

AS(1), {AS(2)}

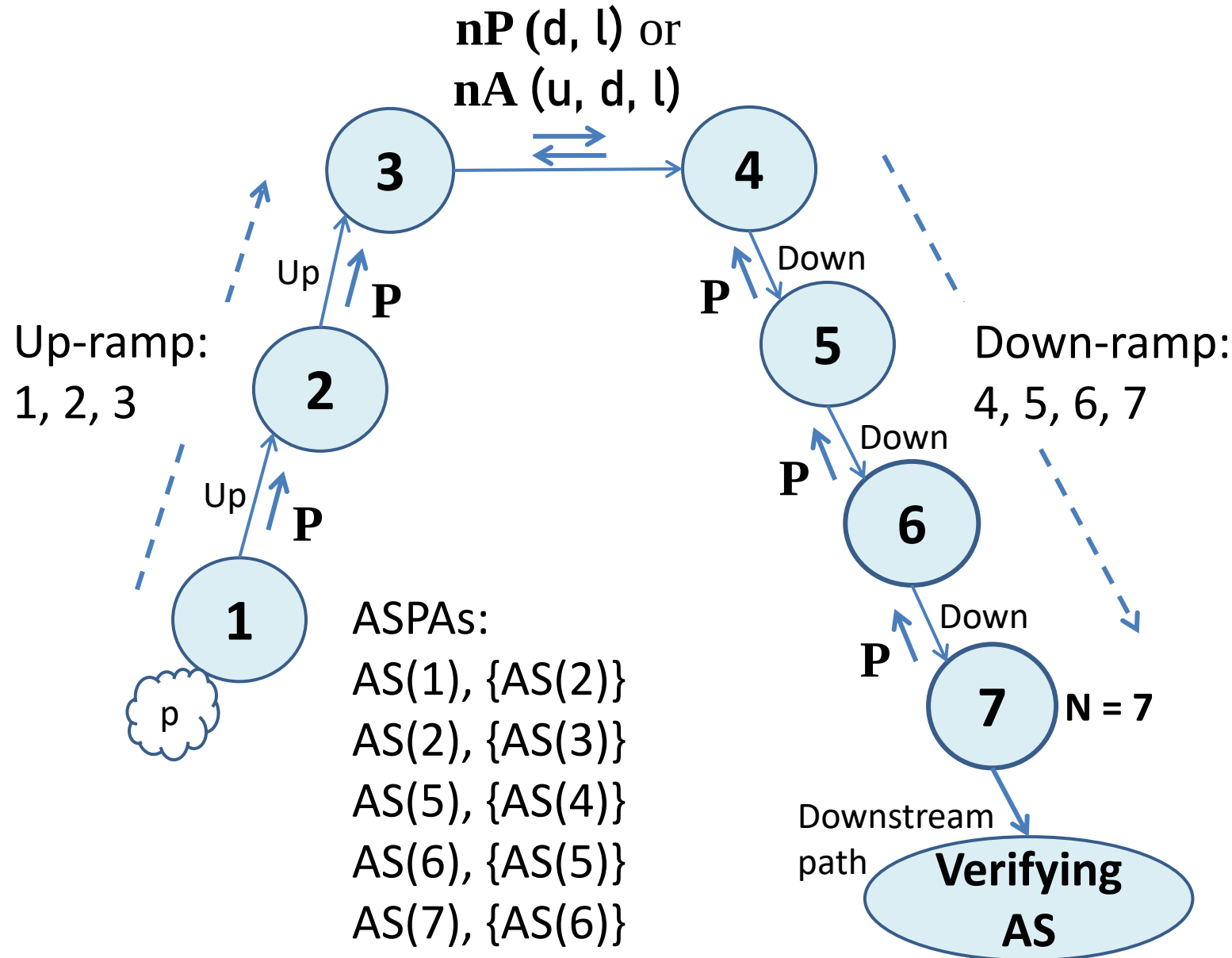
AS(2), {AS(3)}

AS(3), {AS 0}

AS(4) is the leaker and may or may not have an ASPA...

the Verifying AS has local knowledge that AS(4) is its customer

ASPA Verification of Downstream AS Path: **Valid Outcome**



Shortcoming:

The verifying AS cannot tell if AS 4 is faking a link to AS 3

P: Provider

nP: not Provider

nA: no Attestation

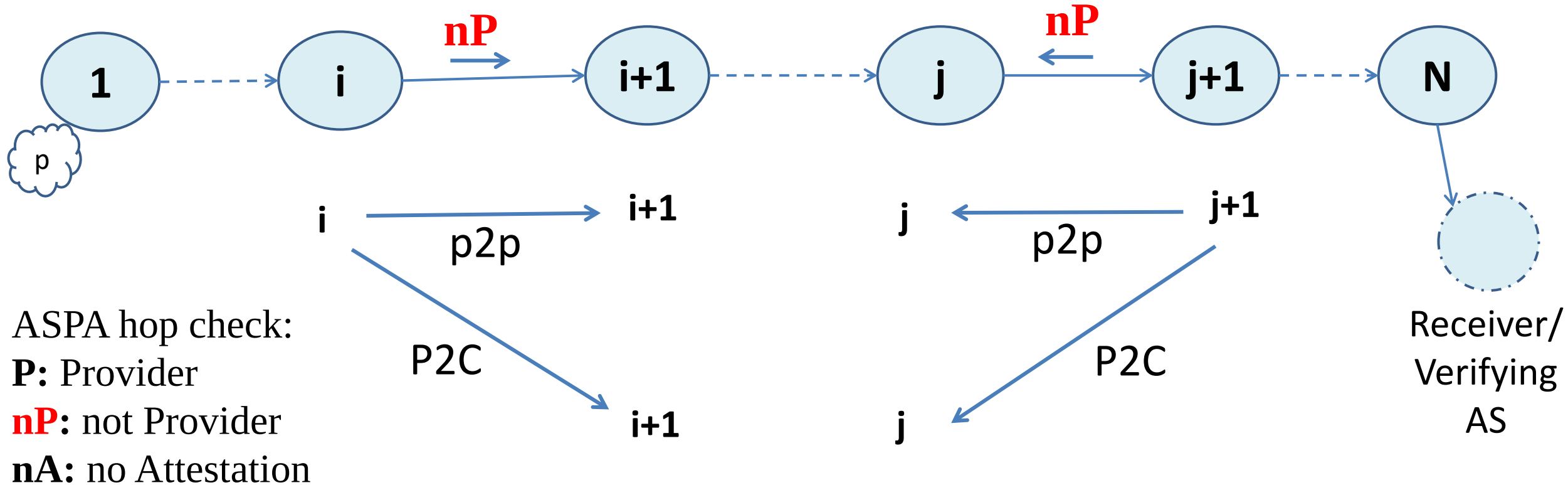
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ASPA Verification of Downstream AS Path:

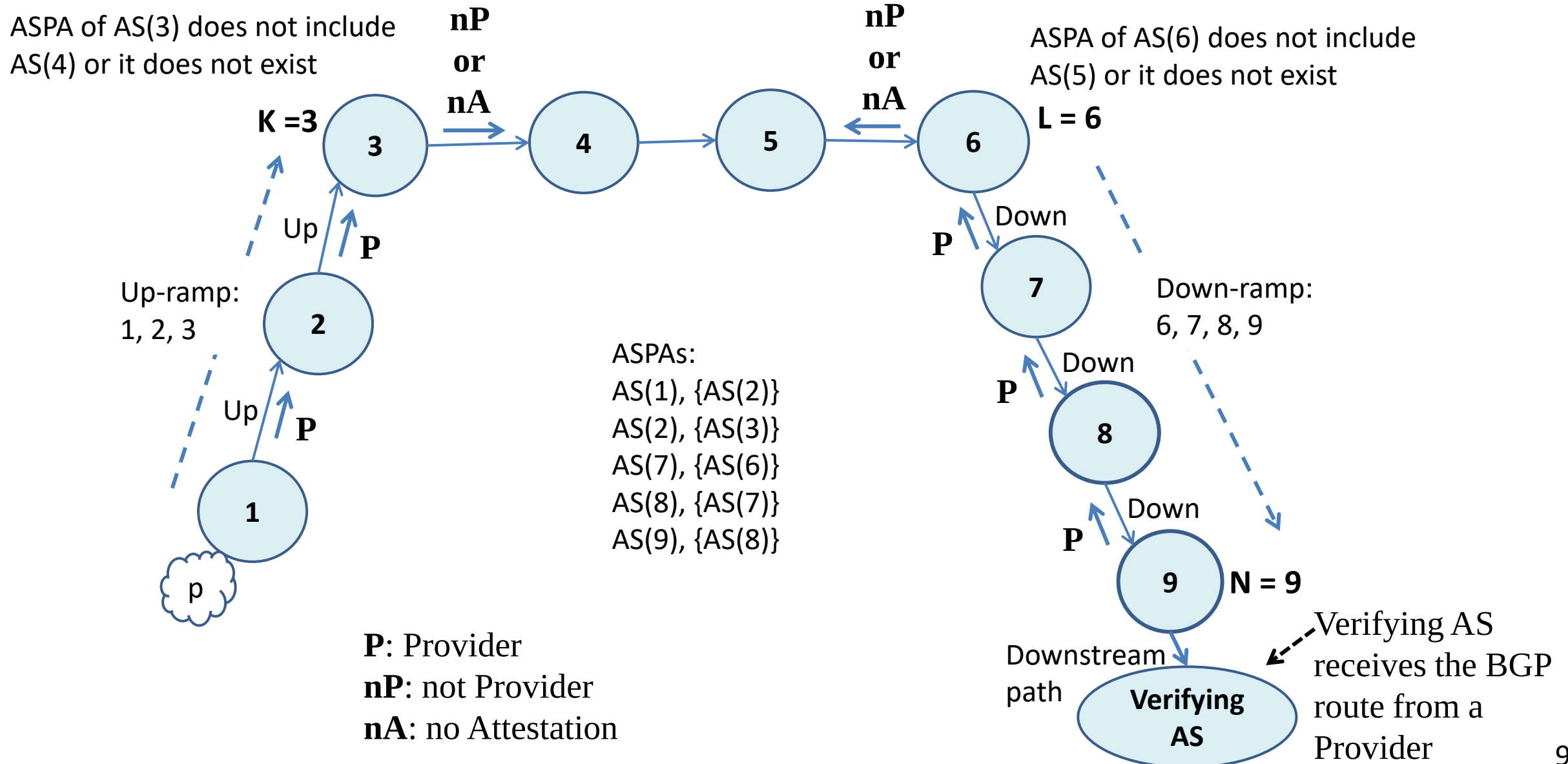
Invalid Outcome



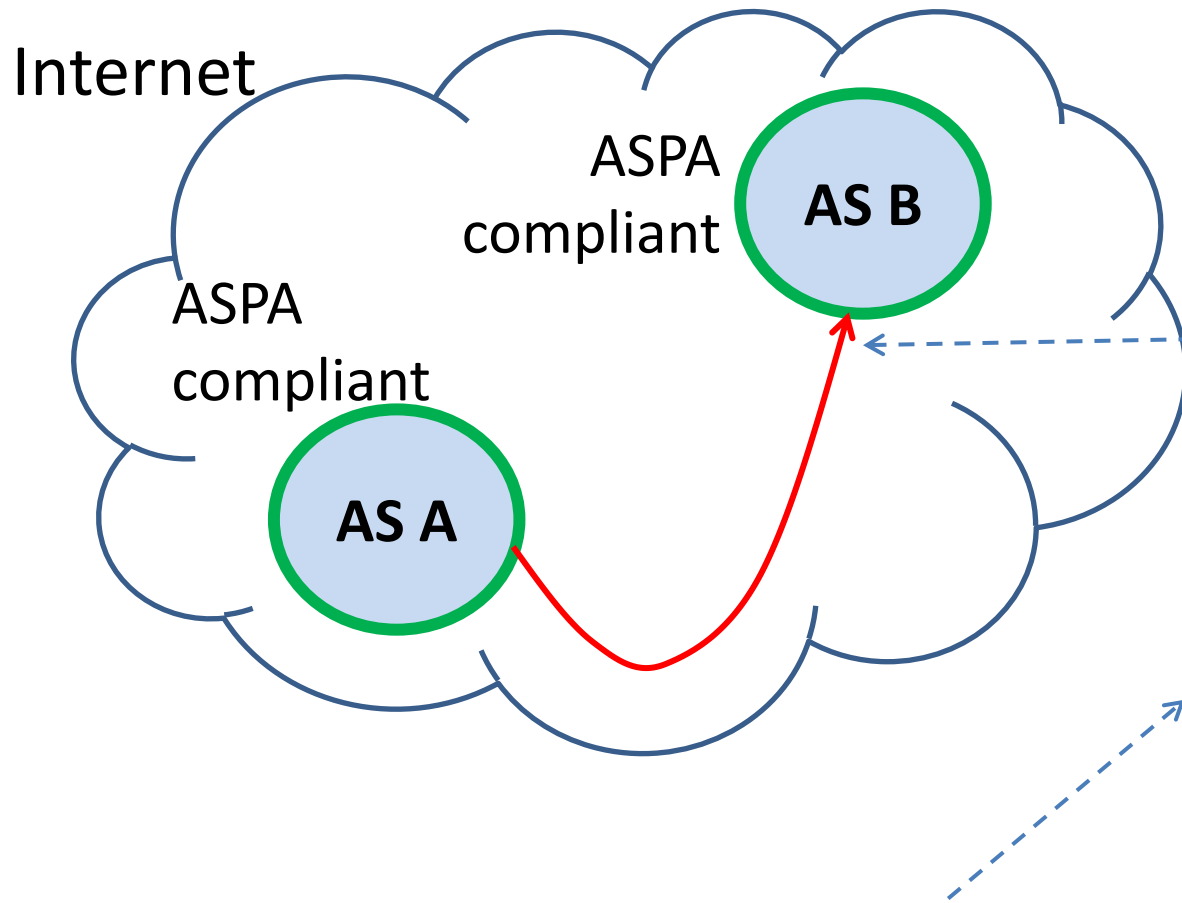
Any two hops in opposite directions are **nP** per ASPA ($j > i$)
(facing each other)

ASPA Verification of Downstream AS Path

(K, L) representation of downstream AS path



ASPA Path Verification: Property 1

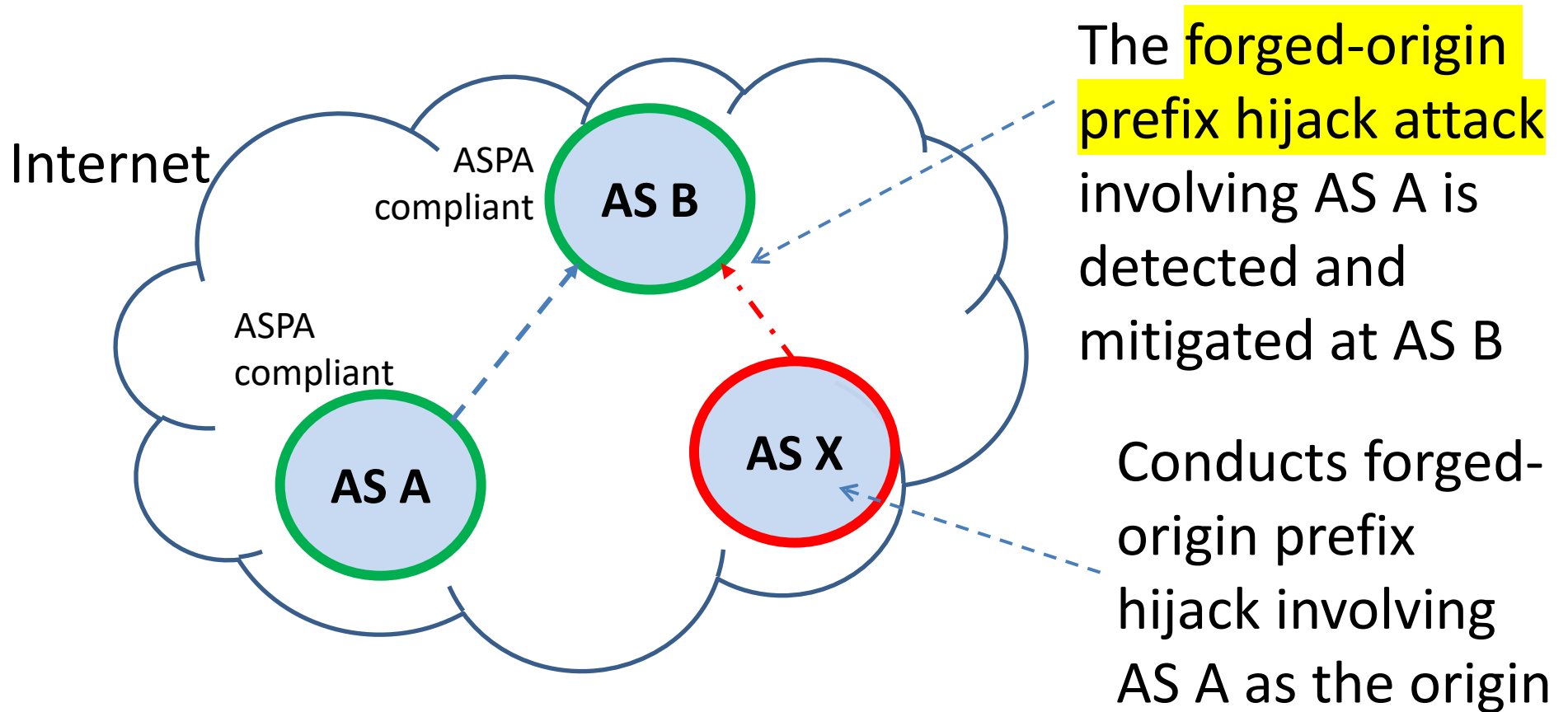


- Only two ASes A and B are doing ASPA
- AS A propagates a route to a customer or lateral peer
- AS B receives the route from a customer or lateral peer
- If the AS_PATH involves a route leak, it is always detected and mitigated at AS B

Corollary of Property 1

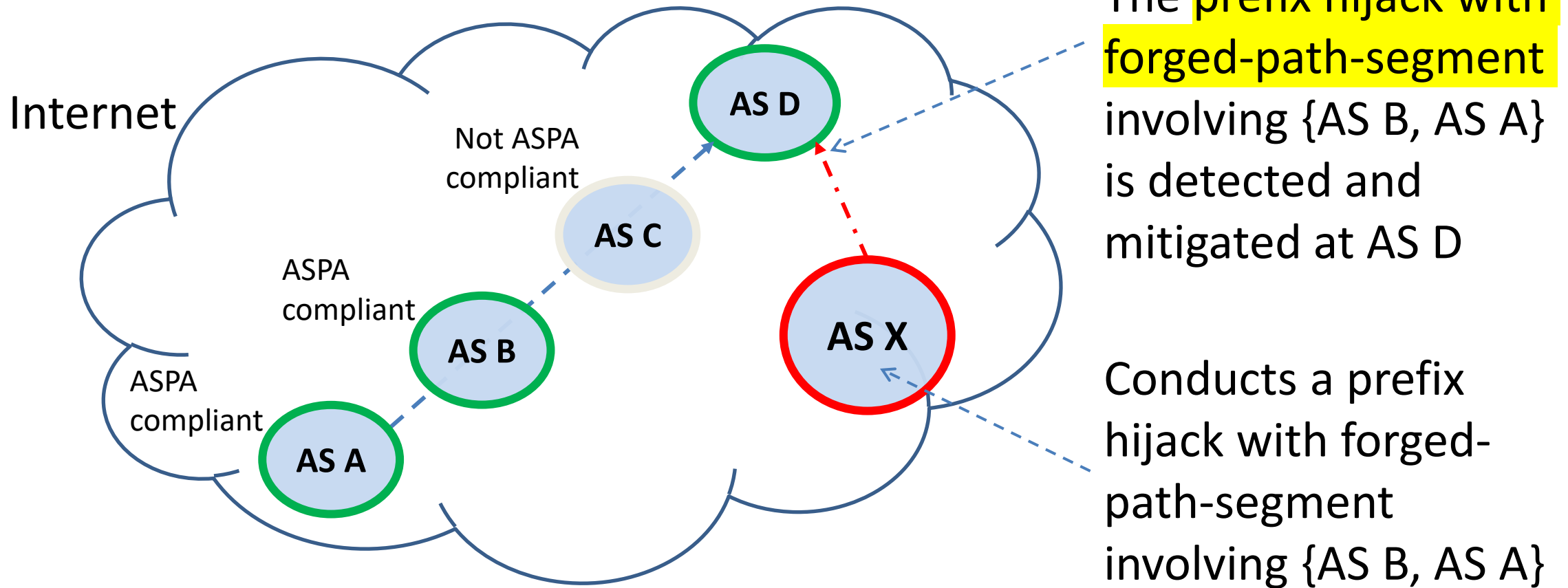
- In effect, if most major ISPs are ASPA compliant, the propagation of route leaks in the Internet will be severely limited.

ASPA Path Verification: Property 2



- Only two ASes A and B are doing ASPA and ROA/ROV
- AS B receives the forged route sent by AS X (attacker) in the upstream direction (from a customer or lateral peer)

ASPA Path Verification: Property 3



- AS B receives the forged route sent by AS X (attacker) in the upstream direction (from a customer or lateral peer)

Autonomous System Relationship Authorization (ASRA) as an Extension to ASPA for Enhanced AS Path Verification

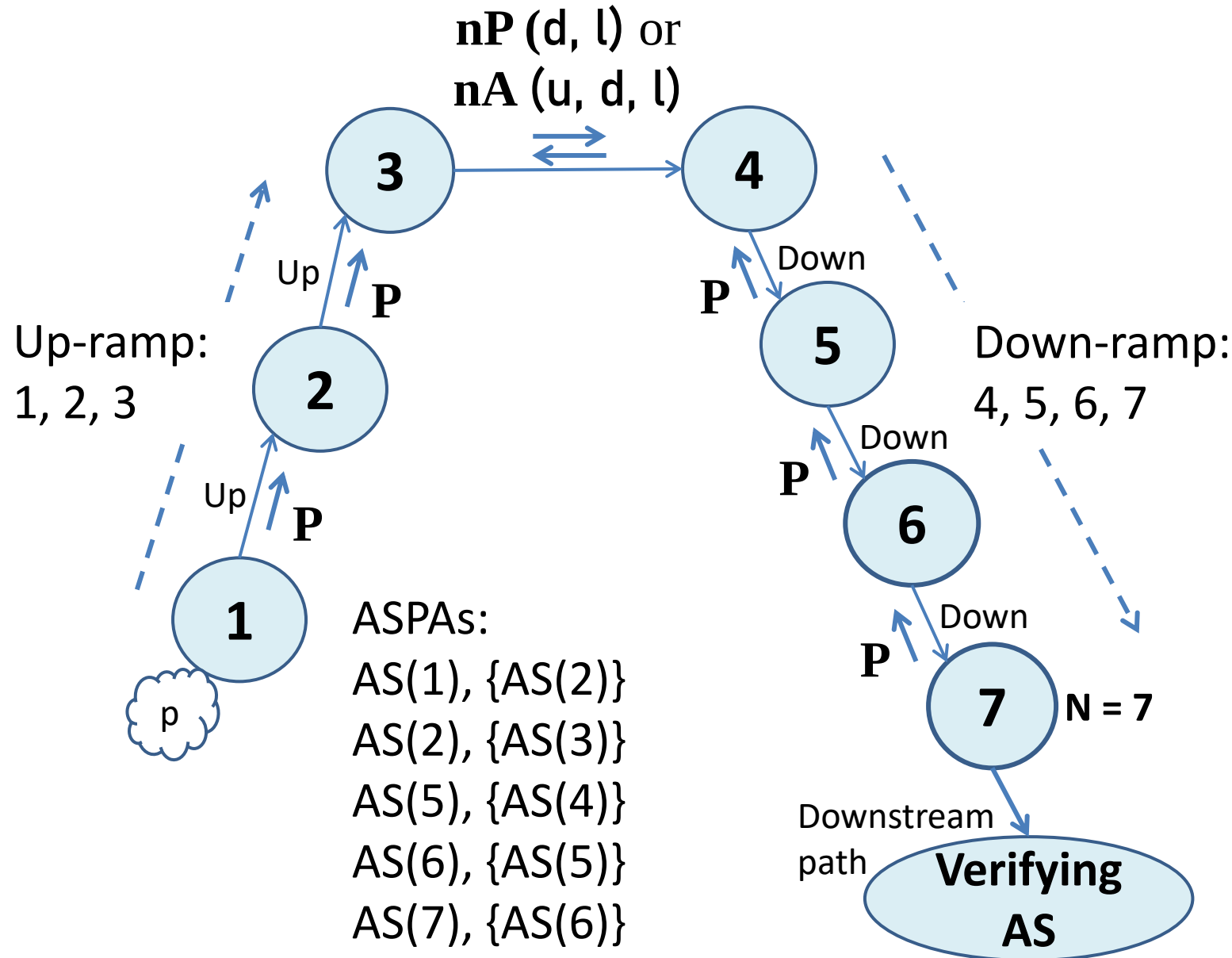
Profile Draft: <https://datatracker.ietf.org/doc/draft-geng-sidrops-asra-profile/>

Verification Draft: <https://datatracker.ietf.org/doc/draft-sriram-sidrops-asra-verification/>

Problem Statement

- ASPA-based AS path verification can detect all route leaks
- It can also detect **forged-origin and forged-path-segment hijacks** when Update is received from a customer or lateral peer
 - **ASPA alone cannot detect these types of hijacks when received from a provider**
 - **These are essentially fake-link (or forged-peering) attacks**
- Goal:
 - Detect and mitigate fake-link attacks received from any direction (i.e., Update received from a provider, customer or lateral peer)
 - **ASPA's route leak detection capability **must** remain intact**

ASPA Verification of Downstream AS Path: **Valid Outcome**



Shortcoming:

The verifying AS cannot tell if AS 4 is faking a link to AS 3

P: Provider

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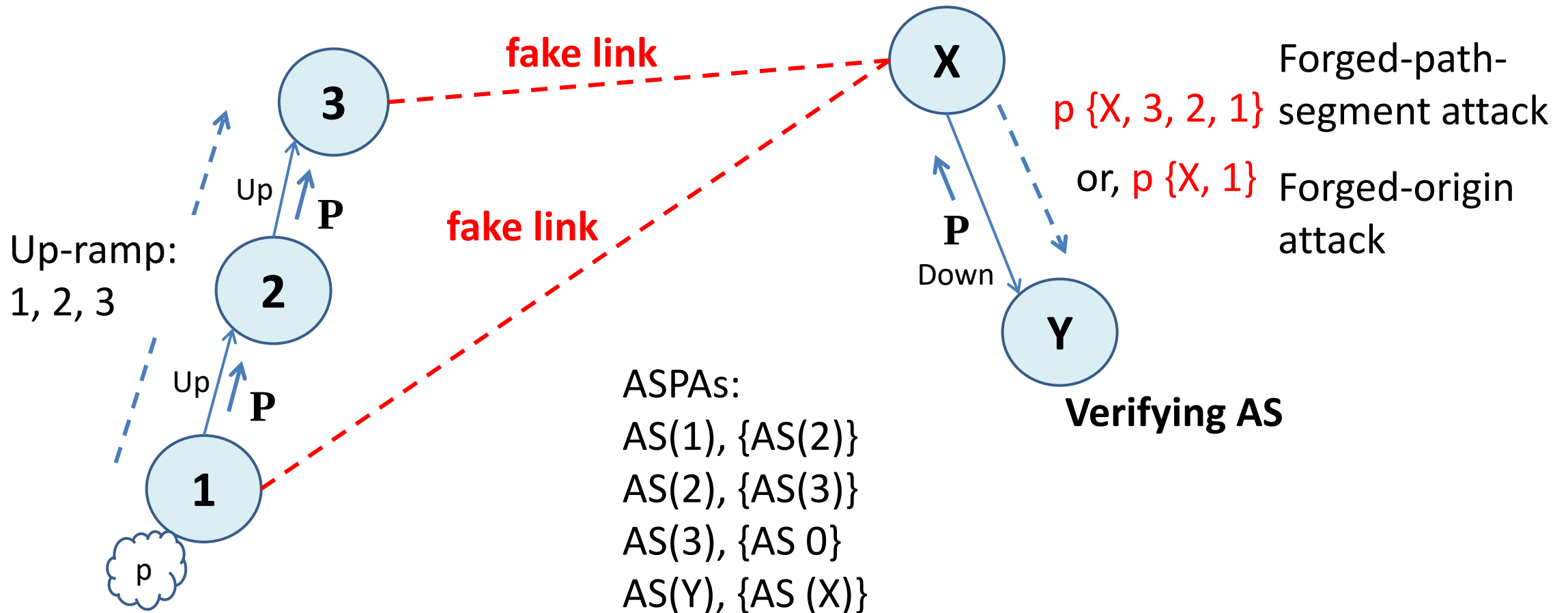
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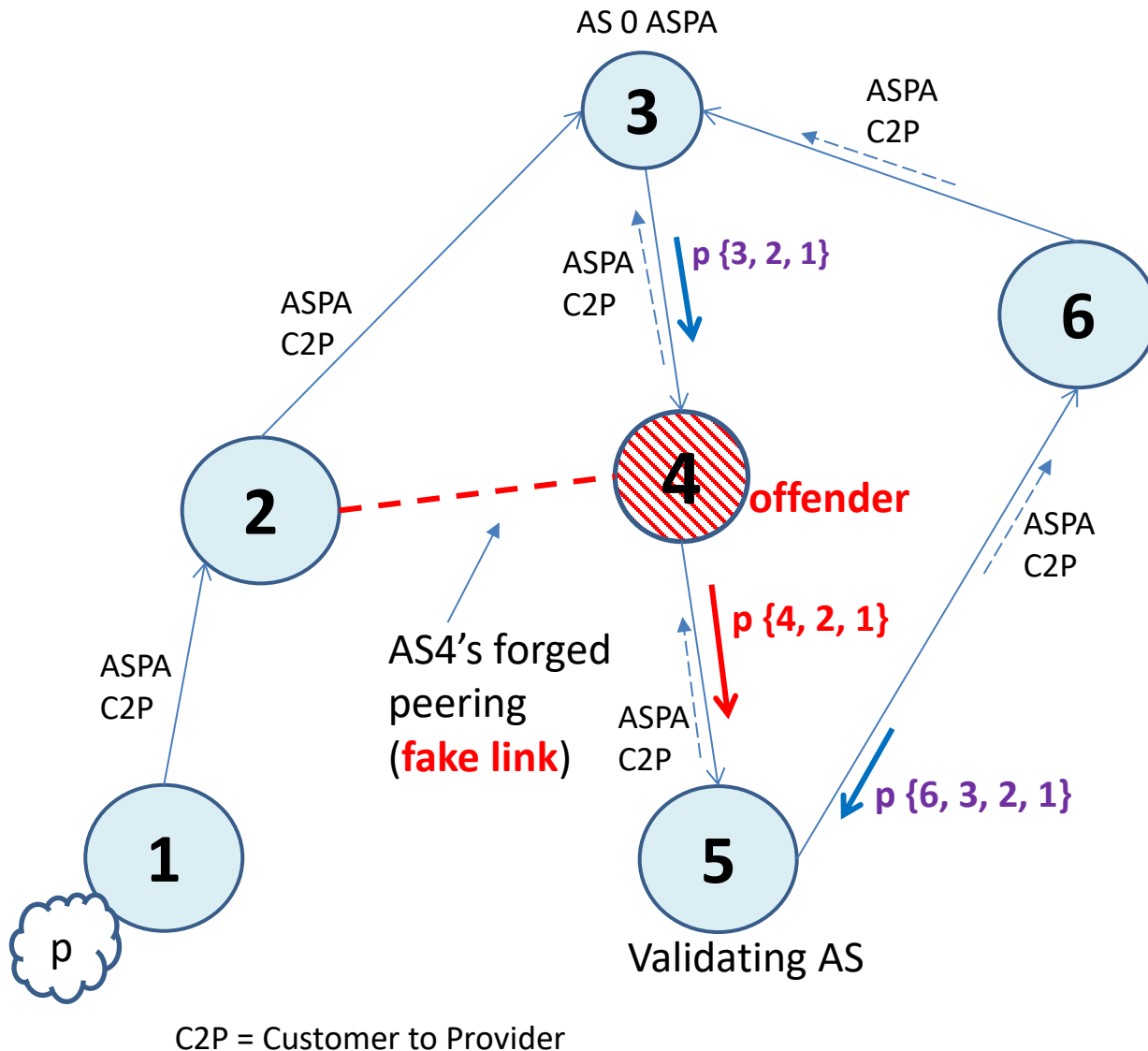
l = Lateral (peer to peer (p2p))

Fake-Link Attack

- ASPA verification not decisively checking the AS hop at the top leaves it open for fake-link attacks



AS_PATH maliciously shortened by a Provider



- All ASes are doing ASPA
- Consider path validation at AS 5
- AS4 (provider) wants AS5 (customer) to prefer its path
- AS4 shortens the AS_PATH
- AS4 does not intend to drop data traffic from AS5 (if it does, that can be addressed only at a higher layer)

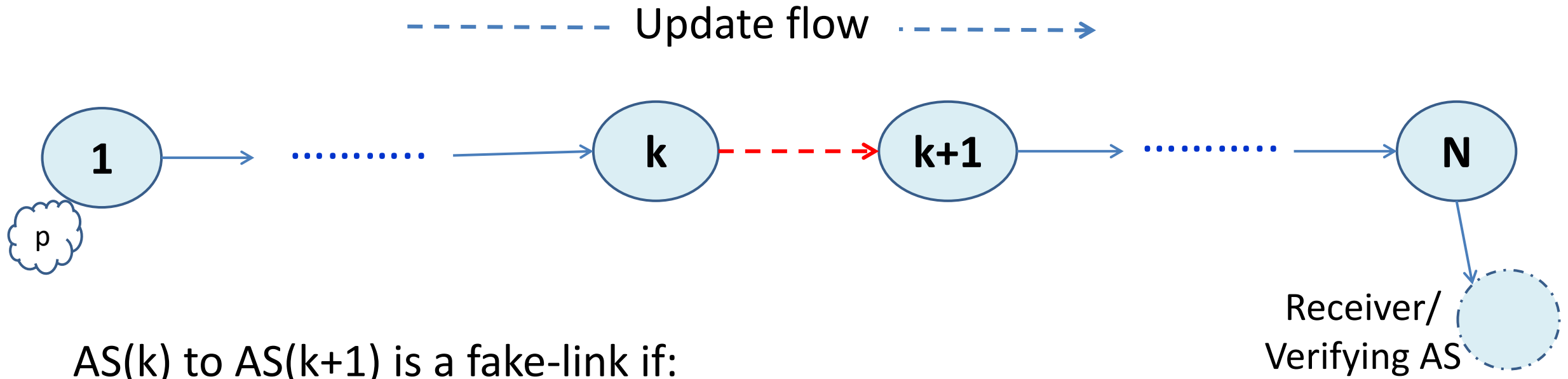
ASPA deficiency: AS4 can shorten the path as shown w/o detection. ASPA verification cannot verify the physical existence of the AS2 to AS4 hop (the presumed inflection hop).

ASRA Subcategories

- Four subcategories of ASRA: ASRA-C, ASRA-LP, ASRA-CLP, ASRA-TIX
- ASRA-C -- register all customers
- ASRA-LP -- register all lateral peers
- ASRA-CLP -- register all customers and lateral peers without differentiation (when subject AS prefers to not disclose its customers explicitly)
- ASRA-TIX is used by a transparent IX (TIX) RS to declare itself as a TIX and it can optionally register all its RS clients

C – Customer; LP – Lateral Peer; CLP – Customer or Lateral Peer

Fake-Link Detection: Algorithm A



AS(k) to AS(k+1) is a fake-link if:

- {ASPA of AS(k) exists and does not include AS(k+1)} AND
- {ASRA of AS(k) exists and does not include AS(k+1) as a customer or lateral peer} AND
- {Either ASPA of AS(k+1) does not exist OR it does not include AS(k)}

BAR-SAV for Source Address Validation and BGP Prefix Filtering

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Authors: **Kotikalapudi Sriram**, Igor Lubashev, and Doug Montgomery

Email: ksriram@nist.gov; ilubashe@akamai.com

IETF ref: <https://datatracker.ietf.org/doc/draft-ietf-sidrops-bar-sav/>

Proposed for NANOG-95, October 2025

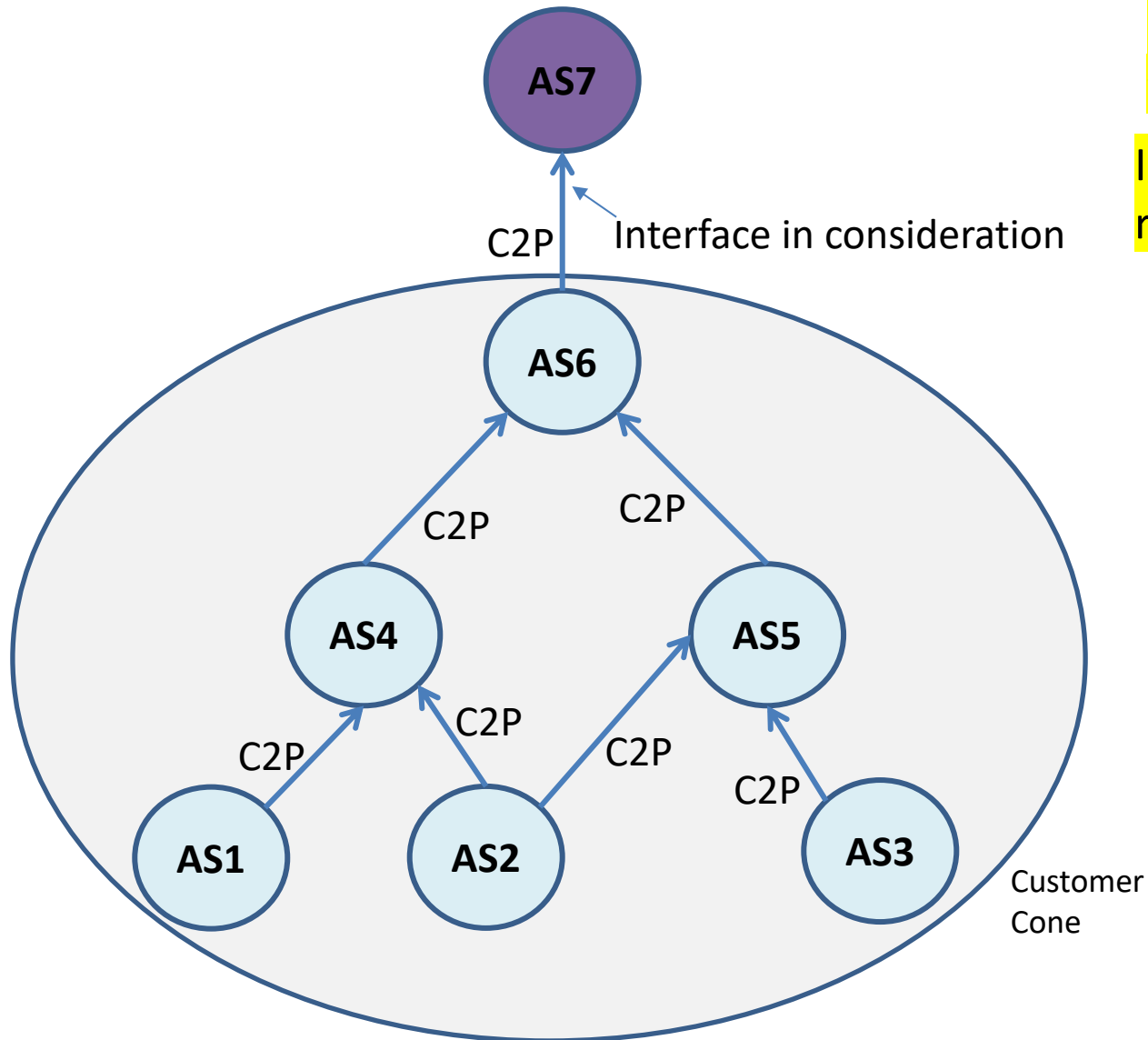
BAR-SAV:

SAV using BGP UPDATE messages, ASPA, and ROA

- A method for mitigating IP address spoofing attacks without inadvertently blocking legitimate traffic in complex networking topologies
- Method for ISP to discover its Customer Cone (CC)
 - What ASes are in the CC?
 - What prefixes are originated in the CC?
 - Useful for ISP BGP Prefix Filtering

SAV Using Only ASPA and ROA (Procedure X)

Construction of Permissible Ingress Prefix List for SAV (at AS7)



When ASPA and ROA adoption is ubiquitous (in the future)

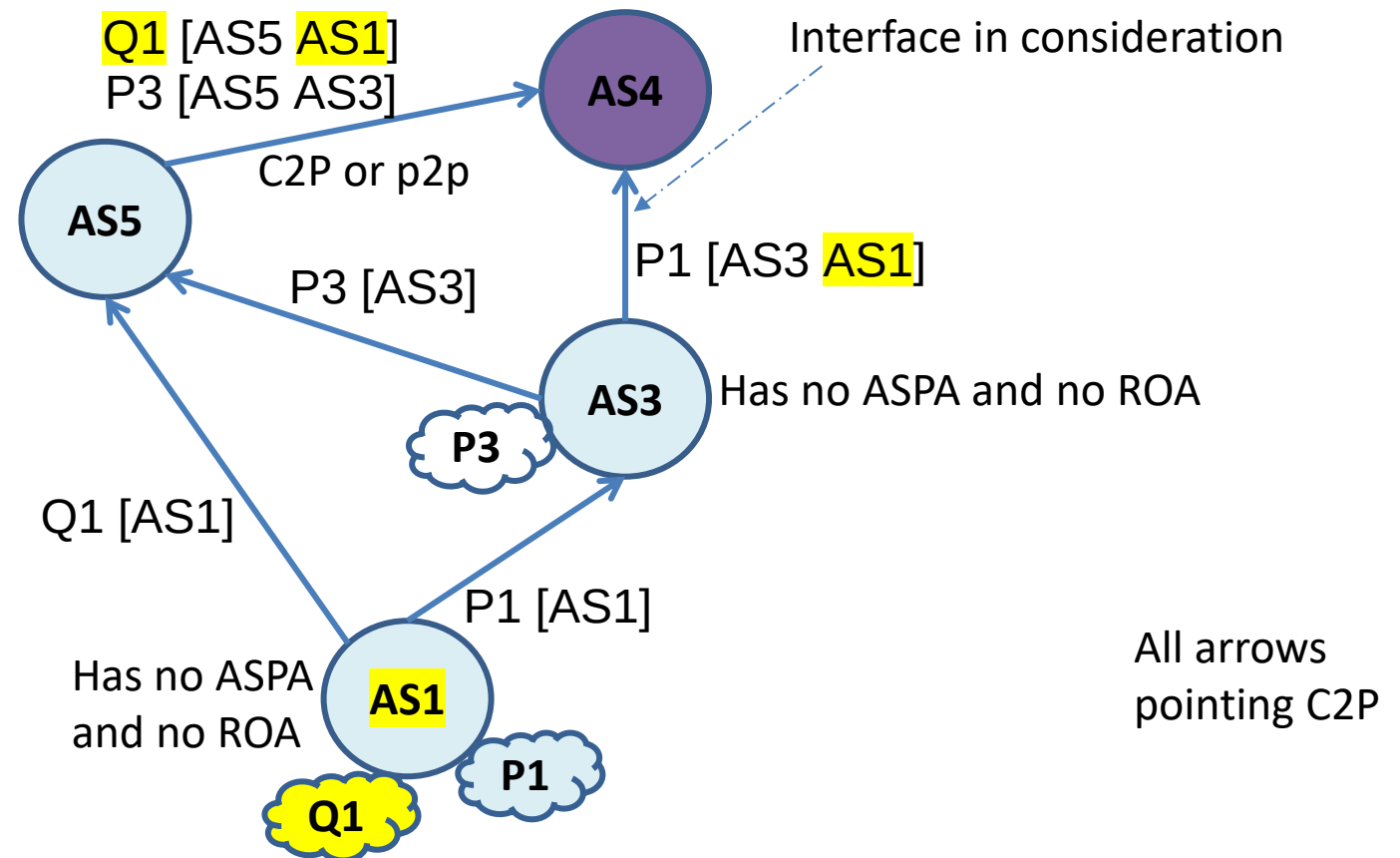
ISP may use Procedure X on customer interfaces if it requires all its customers to register ROAs and ASPAs

- Obtain the set of ASNs in the Customer's customer cone (CC) using ASPAs
- Gather all prefixes in ROAs associated with the ASNs found in Step A. Keep only the unique prefixes.
- The set computed in Step B is the permissible prefix list* for SAV for the interface in consideration.

*Can be used for ISP prefix filtering as well

Enhanced Feasible Path uRPF (EFP-uRPF) [RFC 8704] in a Nutshell

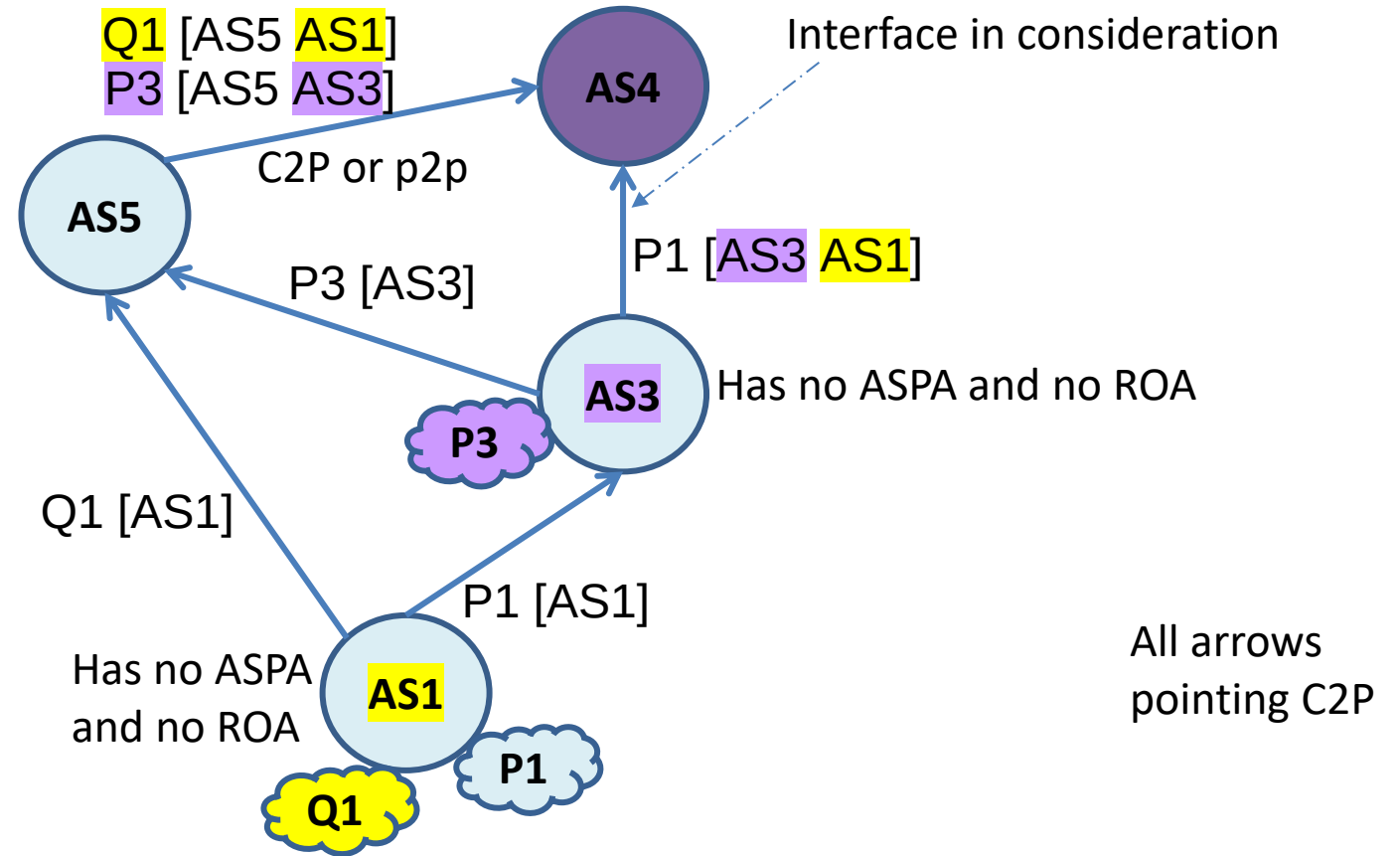
- Careful consideration of asymmetric routing and packet forwarding
- Prefixes P1 and Q1 are captured by Alg. A of RFC 8704
- P3 is not captured
- EFP-uRPF considered only the origin AS in AS_PATHs for correlations
- Did not consider the ASes in the middle to capture more hidden prefixes



EFP-uRPF = Enhanced Feasible Path uRPF

Refined Version of EFP-uRPF [RFC 8704] Incorporated into BAR-SAV

- BAR-SAV makes more efficient use of AS_PATHs for discovering more ASes and prefixes in the CC
- Considers ASes in the middle of the path (e.g., AS3) as well for correlations, not just the origin ASes
- Unlike EFP-uRPF, BAR-SAV captures P3 also



- Much better detection of “Hidden” prefixes in multihoming scenarios by BAR-SAV

How BAR-SAV Works

Finding All ASes and Prefixes in Customer's (or Peer's) Customer Cone
Using BGP Announcements (as seen at AS4), ASPA, and ROA

ASPs:

AS3 {AS4,
AS9}

AS5 {AS1}

AS6 {AS1}

AS8 {AS2}

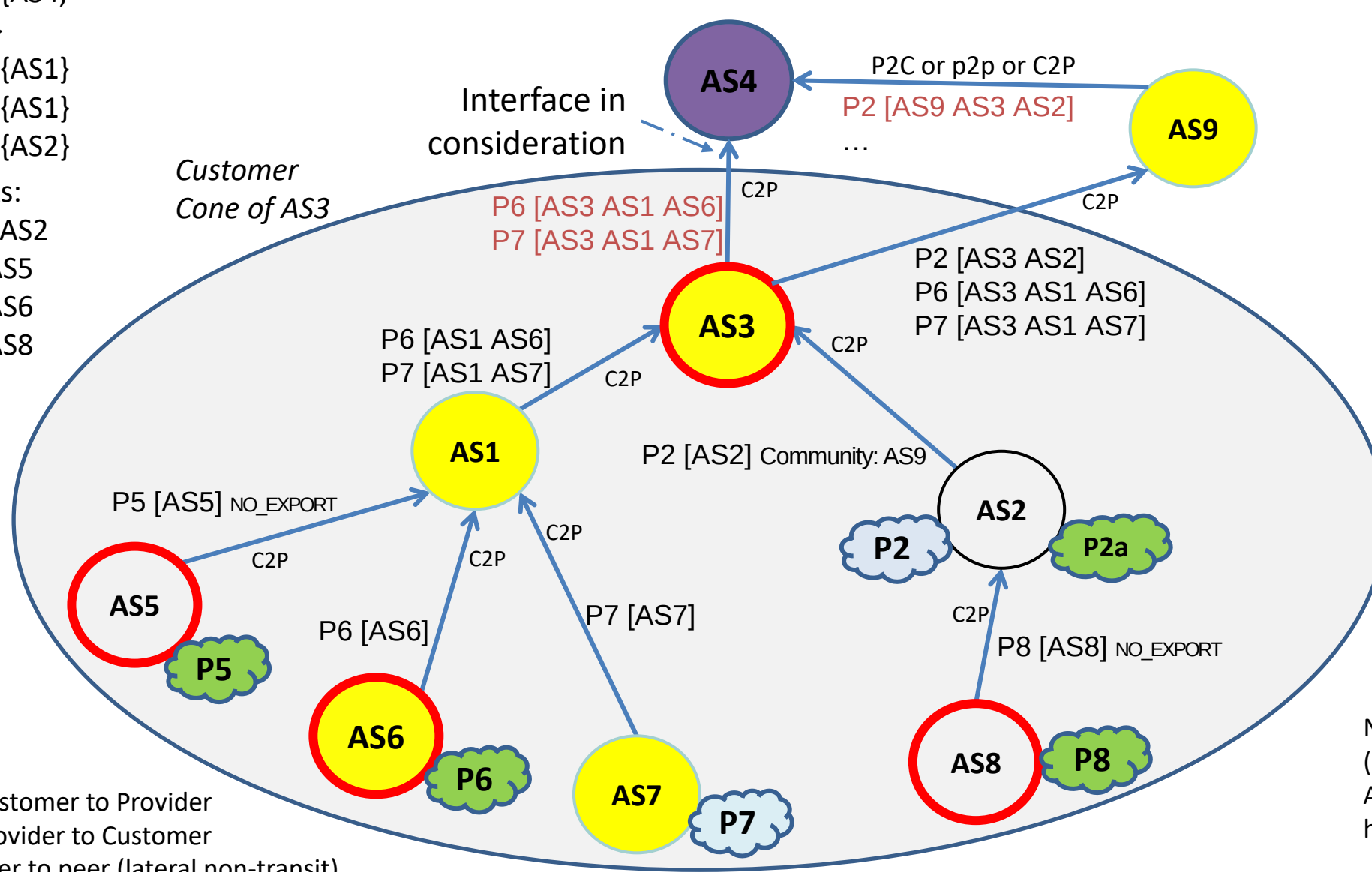
ROAs:

P2a AS2

P5 AS5

P6 AS6

P8 AS8



C2P: Customer to Provider
P2C: Provider to Customer
p2p: peer to peer (lateral non-transit)

Internet Scale Simulations and Performance Modeling

- Collaborative work:
 - Team consisting of Univ. of Connecticut, Hebrew University of Jerusalem, The College of Management Academic Studies, and NIST
- USENIX NSDI 2026 paper on evaluation of SAV solutions (incl. BAR-SAV):
 - N. Scaglione, J. Furuness, Y. Gilad, H. Leibowitz, C. Morris, B. Wang, K. Sriram, and A. Herzberg, “EZ-SAVE: Evaluation of Easy-to-Deploy Source Address Validation Policies,” Proceedings of the 23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI '26), pp. 2247–2265, May 2026. <https://www.usenix.org/system/files/nsdi26-scaglione.pdf>
- Paper on ASPA + ASRA performance – in preparation

NIST Robust Interdomain Routing (RIDR) project page:

<https://www.nist.gov/programs-projects/robust-inter-domain-routing>